

"Explain It Like I'm 5" Computer Architecture Challenge

CS161 - Design and Architecture of Computer Systems

Instructor: Elaheh Sadredini

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Purpose

Your task is to create a short, engaging, 5-10 minute video explaining an important and hard-to-understand concept in computer architecture for a general audience with no prior background in the topic. The goal is to make a complex idea both understandable and interesting through clear explanations, visuals, and real-world connections. Your video will be uploaded to a YouTube channel so it can be beneficial for others.

Your Video Should

- Explain the concept clearly. Start with the big picture, then break it down step by step.
- Use visuals effectively. Diagrams, animations, or real-world analogies that make the topic easier to follow.
- Highlight why the topic matters. Show how it fits into the bigger picture of computer systems or everyday technology.
- Include at least one example. This could be a calculation, a hardware example, or a simulation.
- Be engaging. Use narration, visuals, and energy. Avoid reading text verbatim.
- Turn on your camera (highly encouraged). Speak directly to the audience for part of the video to make it personal and engaging.
- Introduce yourself briefly. Do this at the beginning or end of your video, keep it short.

Guidelines for Creating the Video

Content

Accuracy first, clarity, and a focus on the hardest part of the concept. Avoid jargon, or explain terms in simple language.

Structure

Hook (10-30 sec), Background, Core Explanation, Application, Recap, and your short self-introduction if not already at the start.

Visuals and Tools

Use diagrams, animations, and annotated visuals. Helpful tools include PowerPoint, Canva, Figma, draw.io, OBS Studio, Clipchamp, CapCut, iMovie, and DaVinci Resolve.

Delivery

Narrate at a clear, moderate pace. Vary tone for emphasis, test your audio to minimize noise, and be personable.

Length and Format

5-10 minutes. Submit an MP4 or an unlisted YouTube link. Videos will be compiled on a course YouTube

channel.

Academic Integrity

Use your own narration and explanations. Credit external images or video snippets. Do not copy existing YouTube videos or lectures.

Example YouTube Channels for Inspiration

- 3Blue1Brown: <https://www.youtube.com/@3blue1brown>
- Computerphile: <https://www.youtube.com/@Computerphile>
- ExplainingComputers: <https://www.youtube.com/@ExplainingComputers>
- Kurzgesagt - In a Nutshell: <https://www.youtube.com/@kurzgesagt>
- Veritasium: <https://www.youtube.com/@veritasium>

Evaluation Criteria (Total: 20 points)

Criteria	Points
Content Accuracy - Concepts explained correctly	6
Clarity and Structure - Logical flow, easy to follow	5
Creativity and Engagement - Analogies, visuals, storytelling	4
Relevance and Application - Why the topic matters	3
Technical Quality - Clear audio, readable visuals	2

Topic Selection

You may choose one topic from the list below, or propose your own topic. If you propose your own topic, check with the instructor before getting started to make sure it is appropriately challenging and relevant.

Suggested Topics

Instruction Set Architecture and CPU Basics

1. RISC vs. CISC, why two design philosophies exist
2. How RISC-V is changing the CPU world
3. How ARM chips dominate mobile devices
4. Superscalar execution, doing more than one thing at once
5. SIMD and vector instructions, supercharging parallel tasks
6. What makes Apple's M-series chips special
7. Why older CPUs can still run modern software
8. How instruction decoding works inside a processor

Caches and Memory Speed

1. What a CPU cache is and why it makes computers fast

2. Multi-level caches (L1, L2, L3), how they work together
3. Cache replacement policies (LRU, FIFO) with everyday analogies
4. Inclusive vs. exclusive caches, different ways to organize speed
5. Why cache misses slow down your computer
6. How video game consoles optimize memory access
7. The memory wall, why faster CPUs do not always mean faster computers

Virtual Memory and Storage Illusions

1. How virtual memory lets your computer pretend it has more RAM
2. Page tables, the map inside your CPU memory system
3. Translation Lookaside Buffer (TLB), a speed booster for memory access
4. Demand paging, loading what you need when you need it
5. Memory protection, how programs are kept from interfering
6. Memory fragmentation, when free space is not really usable

Pipelining and CPU Performance

1. What pipelining is and why it speeds up CPUs
2. Data hazards, when instructions get in each other's way
3. Branch prediction, how your CPU guesses the future
4. Speculative execution, doing work before you know you need it
5. Pipeline stalls, when the CPU has to wait
6. Why some CPUs have fewer GHz but are still faster

Parallelism and Multicore Systems

1. Why your CPU has multiple cores
2. GPUs vs. CPUs, why graphics cards excel at parallel work
3. Heterogeneous computing, mixing big and small cores for efficiency
4. Interconnect topologies, how cores talk to each other
5. Amdahl's Law, understanding the limits of parallelism
6. Why gaming PCs love multi-core processors
7. The challenge of parallel programming

Security and Reliability

1. Spectre and Meltdown, how CPUs were tricked into leaking secrets
2. Rowhammer, attacking memory with reads and writes
3. ECC memory, how computers detect and fix errors
4. Secure boot, making sure your system starts safely
5. Side-channel attacks, hacking without touching the system
6. How hardware protects your passwords

7. Why you cannot fully trust delete on a computer

Emerging and Exciting Architectures

1. Processing in memory, doing work where the data lives
2. Domain specific accelerators, hardware built for one job
3. Neuromorphic computing, chips that work like brains
4. Quantum inspired architectures, borrowing ideas from quantum computing
5. Optical computing, using light instead of electricity
6. Why AI chips are designed differently than CPUs
7. 3D stacked processors, building up not out
8. Memristors and new device concepts

Storage and I/O

1. NVMe, why modern SSDs are so fast
2. RAID, combining drives for speed and safety
3. Persistent memory, storage that acts like RAM
4. Shingled magnetic recording, how hard drives pack in more data
5. Tiered storage, keeping hot data close and cold data far
6. How cloud data centers store your files
7. Why SSDs slow down when almost full

Interconnects and Communication

1. PCI Express, the highway inside your computer
2. Network on chip, tiny highways inside processors
3. Chiplet based processors, building CPUs like Lego
4. High Bandwidth Memory (HBM), fast memory close to the CPU
5. 2.5D and 3D integration, stacking chips for performance
6. How computers talk to GPUs
7. USB vs. Thunderbolt, what is the real difference

Advanced Execution Models

1. Dataflow architectures, when instructions run as soon as inputs are ready
2. Vector processors, processing data in wide lanes
3. Decoupled access and execute, splitting memory access from computation
4. Run ahead execution, working ahead while waiting for memory
5. Approximate computing, trading accuracy for speed
6. Out of order execution, why instructions do not run in order
7. How branchless programming speeds things up

Newly Added Fun and General Topics

1. Why your phone CPU is different from your laptop CPU
2. How a smartwatch does so much with little power
3. Integrated vs. dedicated GPUs
4. Why heat is the enemy of fast computers
5. How overclocking works and why it is risky
6. Moore's Law, past, present, and future
7. The history of turbo boost in processors
8. How virtual machines run multiple computers on one
9. Emulators, how old games run on modern PCs
10. How battery life shapes mobile chip design
11. Why low power chips matter for the environment
12. The hidden processors in cars and appliances
13. How cloud gaming works and why latency matters
14. What makes supercomputers fast
15. How processors are tested before being sold
16. Why chips have billions of transistors but still fail sometimes